Memoir on the theory of matrices which was remarkable for containing the first abstract definition of a matrix. He showed that the coefficient arrays studied earlier for quadratic forms and for linear transformations were special cases of his general concept. They simplify our work to a great extent when compared with other straight forward methods which would involve tedious computation. The mathematicians **James Joseph Sylvester** (1814 – 1897), **William Rowan Hamilton** (1805 – 1865), and **Arthur Cayley** (1821 – 1895) played important roles in the development of matrix theory. English mathematician **Cullis** was the first to use modern bracket notation for matrices in 1913. The knowledge of matrices is absolutely necessary not only within the branches of mathematics but also in other areas of science, genetics, economics, sociology, modern psychology and industrial management.

Matrices are also useful for representing coefficients in systems of linear equations. Matrix notations and operations are used in electronic spreadsheet programs on computers, which in turn are used in different areas of business like budgeting, sales projection, cost estimation, and in science, for analyzing the results of an experiment etc.

Interestingly, many geometric operations such as magnification, rotation and reflection through a plane can also be represented mathematically by matrices. Economists use matrices for social accounting, input-output tables and in the study of inter-industry economics. Matrices are also used in communication theory and network analysis in electrical engineering. They are also used in Cryptography.

In this chapter, we now first discuss matrices and their various properties. Then we continue to study determinants, basic properties, minors and their cofactors. Here we now restrict the discussion up to determinants of order 3 only.



Learning Objectives

On completion of this chapter, the students are expected to

- visualise difficult problems in a simple manner in terms of matrices.
- understand different types of matrices and algebra of matrices.
- compute determinant values through expansion and using properties of determinants.
- apply the concepts of matrices and determinants to find the area of a triangle and collinearity of three points.

7.2 Matrices

A **matrix** is a rectangular array or arrangement of entries or elements displayed in rows and columns put within a square bracket [ ].

In general, the entries of a matrix may be real or complex numbers or functions of one variable (such as polynomials, trigonometric functions or a combination of them) or more variables or any other object. Usually, matrices are denoted by capital letters *A, B, C, ...* etc. In this chapter the entries of matrices are restricted to either real numbers or real valued functions on real variables.

General form of a matrix

If a matrix A has m rows and n columns, then it is written as

$A = [a_{ij}]_{m\times n}$, $1 \le i \le m, 1 \le j \le n$.

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